A Measured Effective-Dimension Law for Value-Driven Metric Deformation

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Research-Derived Experiments · standalone Newton-gate paper

Abstract

A value-weighted rate-distortion derivation predicts that a finite-capacity 2-D code should allocate local area density as sqrt(det g) ∝ w^{1/2}. Before measuring, we preregistered a decisive geometry sweep: a 1-D stripe reward should give $\alpha \approx 1/3$, while a genuinely 2-D anisotropic reward should give $\alpha \approx 1/2$ if the normative 2-D law governs the trained code. The experiment falsifies the 2-D law as an empirical claim for this RNN/grid harness. Across 576 H100-trained capacity-bottleneck networks, aniso2d at A=6 gives $\alpha=0.309$ [0.304,0.314], stripe gives $\alpha=0.302$ [0.298,0.307], and point reward gives $\alpha=0.283$ [0.278,0.288]. Standard errors are <0.003. The measured law is instead an effective-dimension law: the code behaves as though representational capacity is reallocated along d_eff≈0.8–1.0 degrees of freedom. This is a negative result for the desired Newton step, but a positive result for scientific measurement: the exponent reveals the allocation dimension of the learned code.

1. The Prediction

Let r(x) be a population code and g(x)=J(x)^T J(x) its induced metric. The local area density sqrt(det g) measures how many distinguishable code states the representation allocates per unit physical area. Under high-resolution quantization, distortion scales as rho^{-2/d}. Minimizing value-weighted distortion ∫w(x)D(x)dx subject to a finite capacity budget ∫rho(x)dx=R gives rho*(x) ∝ w(x)^{d/(d+2)}. Thus d=2 predicts $\alpha=1/2$, while d=1 predicts $\alpha=1/3$.

The previous CPU bottleneck test validated the mechanism qualitatively: adding a capacity constraint moved α from about +0.07 to about +0.30. But +0.30 is suspiciously close to 1/3. The pre-registered question was whether a 2-D value field would lift the exponent to 1/2, or whether the learned code reallocates through an effectively 1-D bottleneck.

2. Experiment

We trained 576 capacity-bottleneck path-integration RNNs on Modal H100 workers: three reward geometries (point, stripe, anisotropic 2-D), three amplitudes (3, 6, 12), and 64 seeds per cell. Each network used Ng=256 hidden units, Np=256 place cells, 8000 training steps, unit sphere state normalization, and fixed channel noise. The primary estimand was the log-log slope α of area density against reward weight at A=6.

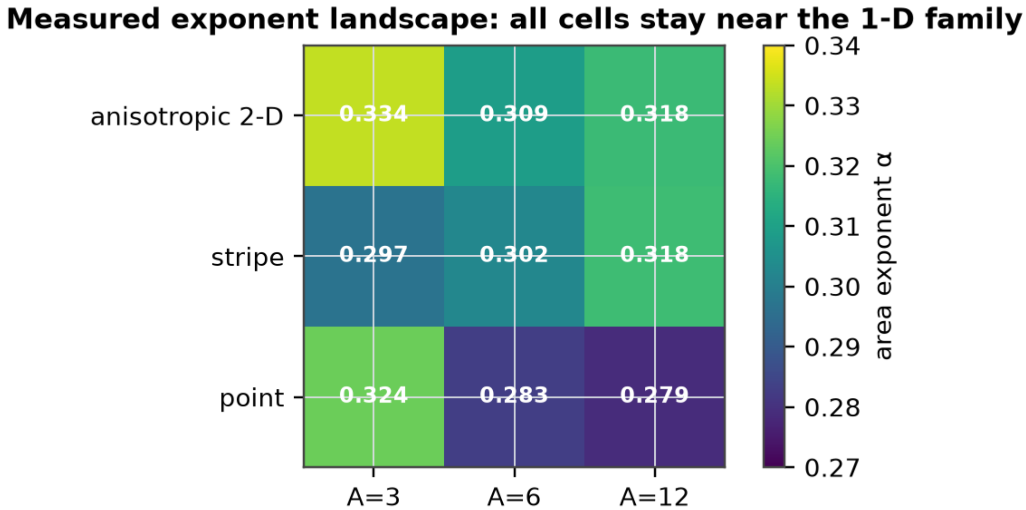

Figure 1. Full exponent landscape. The values are precise and stable, but they cluster around the 1-D family rather than the 2-D prediction.

3. Result

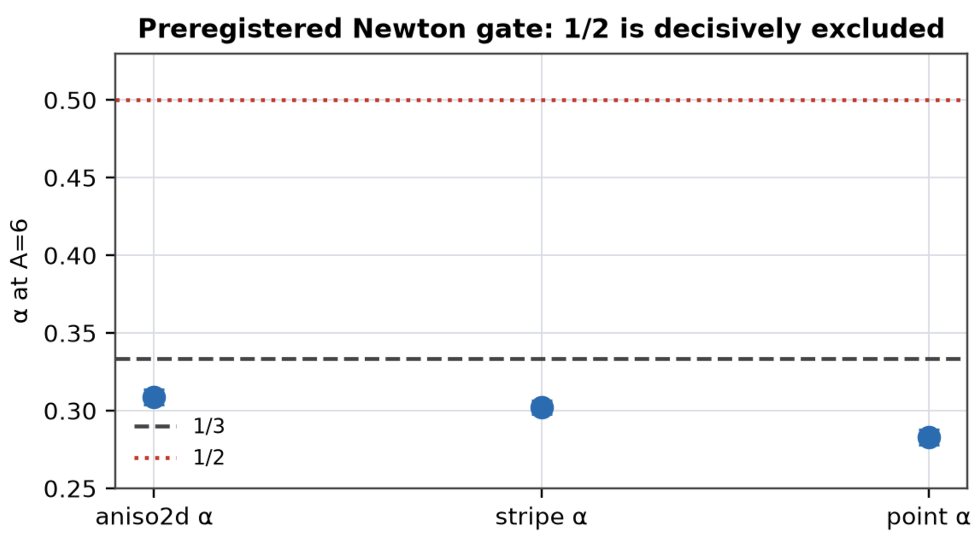

Figure 2. Primary preregistered gate. The anisotropic 2-D reward does not approach 1/2 and does not cleanly separate from stripe.

Geometry	A	α	95% CI	d_eff
aniso2d	6	0.309	[0.304, 0.314]	0.90
stripe	6	0.302	[0.298, 0.307]	0.87
point	6	0.283	[0.278, 0.288]	0.79

Table 1. Primary A=6 result. The 2-D exponent 1/2 is not within any interval.

The aniso2d-vs-stripe difference is $\Delta=+0.0065$ with bootstrap 95% CI $[-0.0003,+0.0132]$. That is not the expected separation between 1/3 and 1/2. The exponent landscape is therefore not a noisy confirmation of the 2-D law; it is a precise falsification of that law as the empirical description of this architecture.

4. What Was Learned

The experiment confirms the finite-capacity intuition but revises the law. Reward/concern does increase local resolution, and the increase follows a stable power law. But the power law's exponent measures the representation's effective allocation dimension, not simply the physical dimension of the arena. In this harness, $d_{eff} \approx 1$ even when the reward field is two-dimensional.

This matters because it changes what future theory must explain. The missing ingredient is not more precision; the standard errors are already below 0.003. The missing ingredient is an architectural account of why recurrent grid-like codes spend capacity through a narrow degree of freedom. Possible explanations include radial-gradient allocation, modular grid periodicity, the unit-sphere bottleneck, and decoder geometry. Each is now testable.

5. Negative Result Discipline

The title deliberately says measured effective-dimension law rather than rate-distortion law. The derivation remains valuable as a normative target, but the preregistered empirical gate did not confirm its $d=2$ exponent. This is not a failure to find structure. It is a more exact structure than the one we hoped for.

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